

The Bounded Multiverse of Living Subjects: A Topological-Complementarity Architecture for Non-Puppet Agency in Interactive Worlds

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Abstract

The phenomenological distinction between a “living subject” and a “puppet” in interactive media is often attributed to graphical fidelity, animation quality, or dialogue volume. This paper argues that the true source of perceived subjectivity is structural rather than visual. Using *Resident Evil 4* as a paradigmatic case, we formalize the notion of a Bounded Hierarchical Complementarity (BHC) topology: a finite location-multiverse in which every entity—NPC, bot, object, environmental feature, or architectural element—is simultaneously a subject, a constraint, and a complement to every other entity. We introduce the Subject Coherence Functional $\mathcal{C}(\mathcal{L})$, the Living-Subject Invariant $\Lambda(\mathcal{L})$, and a Verification, Validation, and Uncertainty Quantification protocol for subjectivity (VVUQ-S). We show that puppet-like behavior arises when the complementarity graph collapses into a star topology centered on the player or when entities become instruments of an external observer. By contrast, living worlds are bounded, connected, hierarchical, non-contradictory, and cyclic. The same architecture appears in cell biology, crystal lattices, ecosystems, architecture, and musical polyphony. We conclude that subjectivity is not a graphical effect but a topological invariant of bounded complementarity.

1 Introduction: The Puppet Problem

In many interactive systems, NPCs feel like puppets or content dispensers. They wait for the player, speak only when approached, and vanish from causal relevance once the player leaves. Their behavior can be written as a function of the player state:

$$f_{\text{NPC}}(\psi_{\text{player}}).$$

Such entities are instruments. They do not inhabit a world; they serve an observer.

By contrast, in *Resident Evil 4*, the village, castle, and island feel like bounded worlds. The Ganados are not merely enemies placed for the player’s amusement. They exist within a village that has its own internal order: houses, paths, forge, church bell, dogs, breakable objects, patrol logic, audio zones, and hierarchical relations. Each entity is both constrained by and constraining to other entities.

The central thesis of this paper is:

An entity appears as a living subject if and only if it belongs to a bounded, hierarchical, non-contradictory complementarity graph in which its purpose is not reducible to service for an external observer.

2 The Bounded Location-Multiverse

Define a Location-Multiverse as a quadruple:

$$\mathcal{L} = (\Omega_{\mathcal{L}}, \partial\Omega_{\mathcal{L}}, \mathcal{E}, \mathcal{H}),$$

where:

- $\Omega_{\mathcal{L}}$ is the internal state space of the location;
- $\partial\Omega_{\mathcal{L}}$ is the hard boundary of the location;
- $\mathcal{E} = \{e_1, e_2, \dots, e_M\}$ is the set of all entities;
- $\mathcal{H}$ is the hierarchical stratification of entities.

The boundary $\partial\Omega_{\mathcal{L}}$ is essential. It turns the location into a closed world. Without a boundary, entities leak into instrumental relations with an outside. With a boundary, entities must define one another internally.

Each entity e_i is described by:

$$\psi_i = (\psi_i^{\text{purpose}}, \psi_i^{\text{constraint}}, \psi_i^{\text{complement}}, \psi_i^{\text{stratum}}).$$

Here:

- ψ_i^{purpose} is the entity's internal reason for existing;
- $\psi_i^{\text{constraint}}$ is the set of constraints it imposes;
- $\psi_i^{\text{complement}}$ is the set of entities it completes and by which it is completed;
- ψ_i^{stratum} is its hierarchical level.

3 Complementarity and Non-Contradiction

Define the Complementarity Operator:

$$\hat{\mathcal{K}}(e_i, e_j) = \left\langle \psi_i^{\text{complement}}, \psi_j^{\text{complement}} \right\rangle - \left\| \psi_i^{\text{purpose}} - \psi_j^{\text{constraint}} \right\|^2.$$

The first term measures mutual completion. The second term penalizes contradiction. A location produces living subjects only if contradictions are suppressed.

Axiom 1 (Non-Contradiction). *For all distinct entities $e_i, e_j \in \mathcal{E}$,*

$$\hat{\mathcal{K}}(e_i, e_j) \geq 0.$$

If $\hat{\mathcal{K}}(e_i, e_j) < 0$, at least one entity is forced into instrumental or contradictory status. It becomes a puppet, a glitch, or a prop.

4 Subject Coherence Functional

We define the Subject Coherence Functional:

$$\mathcal{C}(\mathcal{L}) = \frac{1}{|\mathcal{E}|^2} \sum_{i=1}^M \sum_{j=1}^M \hat{\mathcal{K}}(e_i, e_j) \mathbb{K}[\mathcal{H}(e_i) \neq \mathcal{H}(e_j)] \delta_{\partial\Omega_{\mathcal{L}}}.$$

The indicator encourages cross-stratum complementarity. The confinement term $\delta_{\partial\Omega_{\mathcal{L}}}$ penalizes entities whose purpose references an outside observer.

Theorem 1 (Living Subject Condition). *A location $\mathcal{L}$ produces entities perceived as living subjects if and only if:*

$$\mathcal{C}(\mathcal{L}) > \mathcal{C}_{\text{crit}} > 0$$

and for every entity e_i ,

$$\deg_{\text{complement}}(e_i) \geq 2.$$

An entity with $\deg_{\text{complement}}(e_i) = 0$ is isolated and inert. An entity with $\deg_{\text{complement}}(e_i) = 1$, especially if connected only to the player, is an instrument.

5 The No-Instrument Principle

The mathematical signature of a subject is:

$$\frac{\partial \psi_i^{\text{purpose}}}{\partial \psi_{\text{player}}} = 0.$$

The entity responds to the player but does not exist for the player:

$$\frac{\partial \psi_i^{\text{behavior}}}{\partial \psi_{\text{player}}} \neq 0.$$

In *Resident Evil 4*, a Ganado attacks Leon, but Leon is not the reason the Ganado exists. The Ganado belongs to the village. The player is a perturbation entering a pre-existing bounded multiverse.

6 Complementarity Graph and Living-Subject Invariant

Construct the complementarity graph:

$$G_{\mathcal{L}} = (\mathcal{E}, E_{\mathcal{L}}),$$

where an edge $(e_i, e_j) \in E_{\mathcal{L}}$ exists if and only if

$$\hat{\mathcal{K}}(e_i, e_j) > \epsilon$$

for some threshold $\epsilon > 0$.

Define the Living-Subject Invariant:

$$\Lambda(\mathcal{L}) = \beta_1(G_{\mathcal{L}}) - \beta_0(G_{\mathcal{L}}) + 1 + \sum_{k=1}^K \frac{|\text{edges within } S_k|}{|\text{edges between } S_k \text{ and } S_{k+1}|}.$$

Here:

- $\beta_0(G_{\mathcal{L}})$ is the number of connected components;
- $\beta_1(G_{\mathcal{L}})$ is the number of independent cycles;
- S_k is the k -th hierarchical stratum.

The term $\beta_1 > 0$ indicates feedback cycles. Cycles are essential. They mean that entities mutually sustain one another rather than merely serving a central node.

Corollary 1. *If $\Lambda(\mathcal{L}) \approx 0$, the location is a puppet theater. If $\Lambda(\mathcal{L}) \gg 1$, the location behaves as a living multiverse.*

7 Case Study: Resident Evil 4

7.1 The Village

The village is a bounded multiverse with approximately:

$$|\mathcal{E}| \approx 47.$$

Its hierarchy can be represented as:

$$S_1 \prec S_2 \prec S_3 \prec S_4 \prec S_5 \prec S_6 \prec S_7,$$

where:

- S_1 : environmental substrate;
- S_2 : passive objects;
- S_3 : animals;
- S_4 : NPCs with bounded routines;
- S_5 : adversarial agents;
- S_6 : narrative or parasitic influence;
- S_7 : player as perturbation.

The church bell is especially important. It is not a tool given to the player. It is an event internal to the village. The Ganados stop because the village changes state. The player benefits accidentally.

7.2 The Castle

The castle introduces richer cycles. The Novistadores form a predatory feedback loop:

$$\text{flying units} \rightarrow \text{detection} \rightarrow \text{ground pressure} \rightarrow \text{player displacement} \rightarrow \text{flying units}.$$

This cycle gives $\beta_1(G_{\mathcal{L}}) > 0$. The enemies are not scripts orbiting the player; they are components of a closed predatory ecology.

Table 1: Complementarity structure in the RE4 village.

Entity	Complement	Constraint imposed	Stratum
Village square	Ganado pathing	Defines encounter geometry	S_1
Breakable crates	Player economy	Limits resources	S_2
Dogs	Player attention	Forces split focus	S_3
Blacksmith	Forge, anvil	Anchors defensive position	S_4
Ganado			
Chainsaw Ganado	Ganados, audio	Forces retreat	S_5
Church bell	Village state	Triggers phase transition	S_1
Merchant	Economy	Independent of player	S_4

7.3 The Island

The island introduces military complementarity:

$$\text{infrastructure} \prec \text{command} \prec \text{squads} \prec \text{bio-weapons} \prec \text{player-as-intruder}.$$

Soldiers flank, suppress, communicate, and retreat. Their behavior is internally coordinated, not merely player-triggered.

8 Verification in Sciences

8.1 Cell Biology

A eukaryotic cell is a bounded location-multiverse:

$$\mathcal{L}_{\text{cell}} = (\Omega_{\text{cell}}, \partial\Omega_{\text{cell}}, \mathcal{E}_{\text{cell}}, \mathcal{H}_{\text{cell}}),$$

where the boundary is the cell membrane.

Organelles are not instruments of the nucleus. They are complementary subjects:

$$\text{mitochondria} \rightarrow \text{ATP} \rightarrow \text{ribosomes} \rightarrow \text{proteins} \rightarrow \text{mitochondria}.$$

This is a biological feedback cycle.

Define the Organelle Subjectivity Index:

$$\text{OSI}(e_i) = \frac{\text{number of feedback cycles involving } e_i}{\text{total interactions of } e_i}.$$

If $\text{OSI}(e_i)$ is high, the organelle behaves as a subject. If it is low, the organelle has been instrumentalized.

8.2 Materials Science

A crystal lattice is a bounded multiverse of atoms. Each atom is a subject because its position is determined by mutual constraint.

Define lattice coherence:

$$\mathcal{C}_{\text{crystal}} = 1 - \frac{N_{\text{defects}}}{N_{\text{total}}} \cdot \frac{\sum_{\text{defects}} |\hat{\mathcal{K}}_{\text{violation}}|}{\sum_{\text{all}} |\hat{\mathcal{K}}|}.$$

A perfect crystal has high coherence. A dislocation is a puppet: an atom whose complementarity is broken and whose existence becomes instrumental to stress concentration.

8.3 Ecology

An ecosystem is a bounded multiverse of species. The trophic web is hierarchical and cyclic:

abiotic substrate $\rightarrow$ producers $\rightarrow$ consumers $\rightarrow$ decomposers $\rightarrow$ substrate.

Define the Trophic Subject Coherence:

$$\text{TSC} = \frac{\text{Connectance} \times \text{Modularity}}{\text{Nestedness}}.$$

High TSC indicates a subject-rich ecological multiverse. Low TSC indicates that many species have collapsed into instrumental dependencies.

9 Verification in Arts

9.1 Architecture

A building is a bounded multiverse. In great architecture, every element has purpose, constraint, and complement.

Define Architectural Subject Density:

$$\text{ASD} = \frac{\text{number of elements with } \deg_{\text{complement}} \geq 3}{\text{total number of elements}}.$$

A Gothic cathedral has high ASD because columns, vaults, buttresses, windows, and light shafts mutually define one another. A generic glass office box has low ASD because many elements are interchangeable and instrumental.

9.2 Music

A symphony is a temporally bounded multiverse. Each voice is a subject if it cannot be reduced to accompaniment.

Define Polyphonic Subject Coherence:

$$\text{PSC} = \frac{1}{N_{\text{voices}}} \sum_{i=1}^{N_{\text{voices}}} \frac{H(V_i | V_{-i})}{H(V_i)},$$

where V_i is the i -th voice and V_{-i} denotes all other voices.

If $\text{PSC} \rightarrow 1$, each voice retains irreducible information. If $\text{PSC} \rightarrow 0$, most voices are puppets of a dominant melody or backing track.

10 VVUQ-S Protocol

We propose the Verification, Validation, and Uncertainty Quantification protocol for Subjectivity (VVUQ-S).

10.1 Verification

Given a location $\mathcal{L}$:

1. Enumerate all entities $e_i \in \mathcal{E}$.
2. Construct the complementarity matrix $\hat{\mathcal{K}}(e_i, e_j)$.
3. Check non-contradiction:
$$\forall i \neq j, \quad \hat{\mathcal{K}}(e_i, e_j) \geq 0.$$
4. Construct graph $G_{\mathcal{L}}$.
5. Compute β_0 and β_1 .
6. Require $\beta_0 = 1$.
7. Require $\beta_1 > 0$.
8. Compute $\Lambda(\mathcal{L})$.
9. Require $\Lambda(\mathcal{L}) > \Lambda_{\text{crit}}$.
10. Check:

$$\frac{\partial \psi_i^{\text{purpose}}}{\partial \psi_{\text{external}}} = 0 \quad \forall i.$$

10.2 Validation

Present the system to human observers and measure the Perceived Subjectivity Score (PSS):

$$\text{PSS} = \frac{1}{N} \sum_{k=1}^N \text{score}_k.$$

Expected correlation:

$$r(\text{PSS}, \Lambda) > 0.85.$$

10.3 Uncertainty Quantification

Perturb the boundary, hierarchy, and edge threshold:

$$\partial\Omega_{\mathcal{L}} \mapsto \partial\Omega_{\mathcal{L}} \pm 10\%,$$

$$S_k \cup S_{k+1},$$

$$\epsilon \in [0.01, 0.5].$$

The structure is robust if:

$$\left| \frac{\Delta\Lambda}{\Lambda} \right| < 0.15$$

under boundary perturbation and

$$\left| \frac{\Delta\Lambda}{\Lambda} \right| < 0.20$$

under hierarchical merging.

11 Thermodynamics of Subjectivity

In a puppet system, entropy is low because all entities are determined by the player:

$$H_{\text{puppet}} \approx 0.$$

In a subject system, entropy is high but structured:

$$H_{\text{subject}} \approx \log |\mathcal{E}| - \frac{1}{|\mathcal{E}|} \sum_i \log(\deg_{\text{complement}}(e_i)).$$

Define Subjectivity Free Energy:

$$F_{\text{subj}} = H_{\text{subject}} - T\mathcal{C}(\mathcal{L}),$$

where T is external perturbation, such as player aggression.

A living world remains coherent if it minimizes F_{subj} while maintaining $\mathcal{C}(\mathcal{L}) > \mathcal{C}_{\text{crit}}$.

12 Moral Interpretation

The BHC architecture is the spatial dual of the Moral Pareto Front. Both forbid parasitism.

MPF says:

$$\mathcal{D}(t) = 0,$$

meaning no uncompensated degradation of the environment.

BHC says:

$$\frac{\partial \psi_i^{\text{purpose}}}{\partial \psi_{\text{external}}} = 0,$$

meaning no entity is reduced to an instrument.

Define BHC Honor:

$$\mathcal{H}_{\text{BHC}} = \min_{t \in [0, T]} \left(\frac{\Lambda(\mathcal{L}, t)}{\Lambda(\mathcal{L}, 0)} \cdot \|\Sigma(t)\| \right).$$

Define BHC Pride:

$$\Pi_{\text{BHC}}(t) = \int_0^t \Lambda(\mathcal{L}, \tau) d\tau.$$

Honor is the preservation of complementarity under stress. Pride is the accumulation of coherent complementarity over time.

13 Conclusions

1. The aliveness of NPCs in *Resident Evil 4* is not primarily graphical. It is topological.
2. Living subjects arise in bounded domains where entities are mutually complementary and non-contradictory.
3. Puppets arise when the complementarity graph becomes a star centered on the player.
4. The No-Instrument Condition is the core criterion of subjectivity:

$$\frac{\partial \psi_i^{\text{purpose}}}{\partial \psi_{\text{external}}} = 0.$$

5. The Living-Subject Invariant $\Lambda(\mathcal{L})$ provides a computable diagnostic.
6. The same principle applies to cells, crystals, ecosystems, architecture, and music.
7. Subjectivity is not a bonus feature. It is the invariant of a world that refuses to reduce its inhabitants to tools.

The village does not exist for the player. The player is a temporary perturbation inside a bounded multiverse. That is why the Ganados feel alive. They do not need the player. They have each other, the bell, the forge, the dogs, the houses, the alleys, and the chainsaw. They are a world. The player is only an event that happened to enter it.

A Notation

Table 2: Core notation.

Symbol	Meaning
$\mathcal{L}$	Bounded location-multiverse
$\Omega_{\mathcal{L}}$	Internal state space
$\partial\Omega_{\mathcal{L}}$	Boundary of location
$\mathcal{E}$	Set of entities
ψ_i	State vector of entity e_i
$\hat{\mathcal{K}}(e_i, e_j)$	Complementarity operator
$\mathcal{C}(\mathcal{L})$	Subject coherence functional
$G_{\mathcal{L}}$	Complementarity graph
β_0, β_1	Betti numbers
$\Lambda(\mathcal{L})$	Living-subject invariant
$\mathcal{H}_{\text{BHC}}$	Honor under stress
Π_{BHC}	Accumulated pride
$\deg_{\text{complement}}(e_i)$	Complementarity degree

B Pseudocode for BHC Verification

Listing 1: BHC verification algorithm.

```
1 ALGORITHM: BHC_Verify(L)
2
3 INPUT:
4   L = bounded location
5   E = entities
6   dOmega = boundary
7   H = hierarchy
8
9 OUTPUT:
10  PASS / FAIL
11  Lambda score
12  diagnostic report
13
14 1. Build complementarity matrix  $K[i][j]$ .
15 2. If any  $K[i][j] < 0$ :
16     return FAIL: contradiction detected.
17 3. Build graph  $G_L$  from  $K$  using threshold epsilon.
18 4.  $\beta_0$  = number_of_connected_components( $G_L$ ).
19 5.  $\beta_1$  = number_of_independent_cycles( $G_L$ ).
20 6. If  $\beta_0 > 1$ :
21     return FAIL: disconnected complementarity.
22 7. If  $\beta_1 == 0$ :
23     return FAIL: no feedback cycles.
24 8. Compute Lambda.
25 9. If  $\text{Lambda} < \text{Lambda\_crit}$ :
26     return FAIL: insufficient subjectivity.
27 10. For each entity  $e_i$ :
28      $\text{deg} = \text{complementarity\_degree}(e_i)$ 
29     if  $\text{deg} < 2$ :
30         flag  $e_i$  as puppet risk.
31     if purpose depends on external observer:
32         flag  $e_i$  as instrument risk.
33 11. If flags exist:
34     return CONDITIONAL PASS.
35 12. Else:
36     return PASS.
```

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